

The Hellings-Downs curve for a Gaussian source

Viacheslav Chernosov

April 15, 2026

This paper is devoted to the derivation of the Hellings-Downs curve for the case of a Gaussian source of the Gravitational Wave Background.

1 Problem Statement

Consider the spherical coordinate system (ϕ, θ, r) , where ϕ is longitude, θ is colatitude and r is radius. Then the orthonormal basis in the cartesian coordinate system will look like:

$$\hat{\Omega} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta), \quad (1)$$

$$\hat{n} = (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta), \quad (2)$$

$$\hat{m} = (\sin \phi, -\cos \phi, 0), \quad (3)$$

where $(\hat{\Omega}, \hat{n}, \hat{m}) \equiv (\hat{r}, \hat{\theta}, -\hat{\phi})$. Since $\bar{r} = r \cdot \hat{r}$, and the point on the unit sphere has the value $r = 1$, we can describe the unit sphere with a single vector $\hat{\Omega} = \hat{r}$.

We also know that gravitational waves have two polarization, which are called "plus" and "cross", and are described by the corresponding tensors:

$$e_{\alpha\beta}^+(\hat{\Omega}) = \hat{m}_\alpha \hat{m}_\beta - \hat{n}_\alpha \hat{n}_\beta, \quad (4)$$

$$e_{\alpha\beta}^\times(\hat{\Omega}) = \hat{m}_\alpha \hat{n}_\beta - \hat{n}_\alpha \hat{m}_\beta \quad (5)$$

Then, frequency shift of pulsar radiation $z = \Delta\nu/\nu_0$ in Fourier space:

$$\tilde{z}(f, \hat{\Omega}) = \left(e^{-i2\pi f L(1+\hat{\Omega} \cdot \hat{p})} - 1 \right) h(f, \hat{\Omega}) F_{\hat{p}}(f, \hat{\Omega}), \quad (6)$$

where $\hat{p}$ is the unit vector in the direction of the pulsar, L is the distance to the pulsar, $h(f, \hat{\Omega})$ is spectrum of gravitational waves and $F_{\hat{p}}(\hat{\Omega})$ is pulsar response:

$$F_{\hat{p}}(\hat{\Omega}) = e_{\alpha\beta}(\hat{\Omega}) \frac{1}{2} \frac{\hat{p}^\alpha \hat{p}^\beta}{1 + \hat{\Omega} \cdot \hat{p}}. \quad (7)$$

Here we use a complex representation of the polarization tensor and spectrum, and in the case where the GWB is unpolarized, we have:

$$h(f, \hat{\Omega}) = h_+(f, \hat{\Omega}) + i \cdot h_\times(f, \hat{\Omega}) \quad (8)$$

$$e_{\alpha\beta}(\hat{\Omega}) = e_{\alpha\beta}^+(\hat{\Omega}) + i \cdot e_{\alpha\beta}^\times(\hat{\Omega}) \quad (9)$$

Since the Hellings-Downs curve relates correlations between signals from two pulsars, we defined the Hellings-Downs curve as:

$$\Gamma[\mathcal{P}](\hat{p}, \hat{q}) = \frac{3}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot \mathcal{P}(\hat{\Omega}), \quad (10)$$

where $\mathcal{P}(\hat{\Omega})$ is the power spectrum on the unit sphere, and with the frequency power spectrum $\mathcal{H}(f)$ defined as:

$$|h(f, \hat{\Omega})|^2 = \mathcal{H}(f) \cdot \mathcal{P}(\hat{\Omega}), \quad (11)$$

and $\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q})$ is the integral kernel, which is defined as:

$$\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) = \mathcal{R} \left[F_{\hat{p}}^*(\hat{\Omega}) \cdot F_{\hat{q}}(\hat{\Omega}) \right]. \quad (12)$$

And in our case, we assume that the power spectrum is Gaussian and looks like this:

$$\mathcal{P}(\hat{\Omega}) = \exp \left[-\frac{(\hat{\Omega} - \hat{\Omega}_0)^2}{2\sigma^2} \right], \quad (13)$$

where $\hat{\Omega}_0$ is the center of the source and σ is the width of the Gaussian. Converting the numerator:

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = \hat{\Omega}^2 + \hat{\Omega}_0^2 - 2\hat{\Omega} \cdot \hat{\Omega}_0 = 1 + 1 - 2\hat{\Omega} \cdot \hat{\Omega}_0 = 2(1 - \hat{\Omega} \cdot \hat{\Omega}_0),$$

we have:

$$\mathcal{P}(\hat{\Omega}) = \exp \left[-\frac{2(1 - \hat{\Omega} \cdot \hat{\Omega}_0)}{2\sigma^2} \right],$$

$$\mathcal{P}(\hat{\Omega}) = \exp \left[-\frac{1}{\sigma^2} \right] \cdot \exp \left[\frac{\hat{\Omega} \cdot \hat{\Omega}_0}{\sigma^2} \right].$$

Let $\kappa = 1/\sigma^2$, and as a result we have:

$$\mathcal{P}(\hat{\Omega}) = \exp(-\kappa) \cdot \exp(\kappa \hat{\Omega} \cdot \hat{\Omega}_0). \quad (14)$$

And to find the Hellings-Downs curve for a Gaussian source, we need to calculate:

$$\Gamma^{(g)}(\hat{p}, \hat{q}) = \frac{3e^{-\kappa}}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot e^{\kappa \hat{\Omega} \cdot \hat{\Omega}_0} \quad (15)$$